

SAT Math · Complete Question Set

275 questions · 6 sets

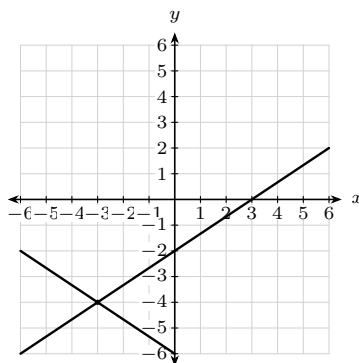
Name _____

Class _____

Date _____

SET 1 · MODULE 1

1. The graph of a system of two linear equations is shown. The solution to the system is (x, y) . What is the value of y ?



- (A) -5 (B) -4
(C) 2 (D) 3

2. The function f is defined by $f(x) = 15x + 10$. What is the value of $f(x)$ when $x = 2$?

- (A) 27 (B) 30
(C) 40 (D) 50

3. In the xy -plane, the graph of $y = f(x)$ is a line that passes through $(0, 21)$ and has a slope of 4. Which equation defines f ?

- (A) $f(x) = 21x$ (B) $f(x) = 21x + 4$
(C) $f(x) = 4x$ (D) $f(x) = 4x + 21$

4. The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which equation represents this relationship?

x	0	1	2
y	22	23	24

- (A) $y = x + 22$ (B) $y = 24x$
(C) $y = 24x + 2$ (D) $y = 22x$

5. One of the two equations in a system of linear equations is given by $y = \frac{3}{4}x + 2$. The system has infinitely many solutions. If the second equation is $y = mx + b$, where m and b are constants, what is the value of b ?

- (A) -2 (B) 0
(C) 1 (D) 2

6. The relationship between x and y is modeled by an exponential equation. When $x = 0$, the value of y is 90, and for every increase in x by 1, the corresponding value of y increases by 20% of its previous value. Which equation represents this relationship?

(A) $y = 90(1.20)^x$

(B) $y = 90(1.02)^x$

(C) $y = 20(1.90)^x$

(D) $y = 20(1.09)^x$

7. The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

Task	Time (min)
1	7
2	9
3	10
4	12
5	12

(A) 8

(B) 9

(C) 10

(D) 11

8. The function h is defined by $h(x) = 7|x|$. What is the value of $h(-2)$?

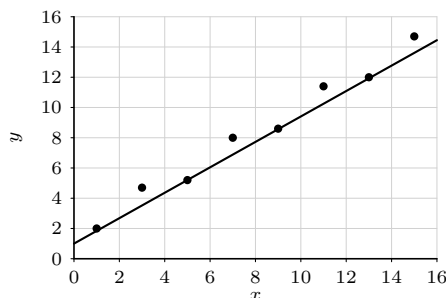
(A) -14

(B) 5

(C) 9

(D) 14

9. Which of the following is closest to the slope of the line of best fit shown?



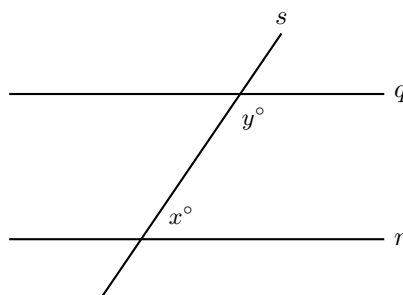
(A) -11.19

(B) -0.84

(C) 0.84

(D) 11.19

10. In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 3x + 16$, what is the value of x ?



(A) 33

(B) 41

(C) 49

(D) 59

11. Raheem bought 9 shirts that were each the same price. He used a coupon for \$54 off the entire purchase. The cost for the entire purchase after using the coupon was \$108. What was the original price, in dollars, for 1 shirt?

Student-produced response: enter your answer as a number.

12. Line t in the xy -plane has a slope of $-\frac{5}{3}$ and passes through the point $(9, 4)$. Which equation defines line t ?

(A) $y = \frac{5}{3}x$

(B) $y = 9x + 4$

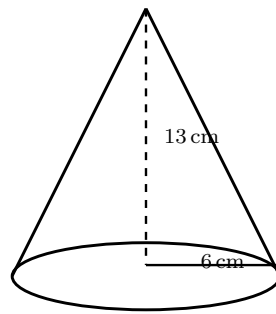
(C) $y = -\frac{5}{3}x + 4$

(D) $y = -\frac{5}{3}x + 19$

13. $(x - 16)(x - 12)(x + 7)(x + 13) = 0$. What is a positive solution to the given equation?

Student-produced response: enter your answer as a number.

14. A right circular cone has a height of 13 centimeters (cm) and a base with a radius of 6 cm. What is the volume, in cubic centimeters, of this cone?



(A) 26π

(B) 78π

(C) 156π

(D) 468π

15. A linear model estimates the population of a city from 1995 to 2019. The model estimates the population was 55 thousand in 1995, 217 thousand in 2015, and x thousand in 2019. To the nearest whole number, what is the value of x ?

Student-produced response: enter your answer as a number.

16. $y = 500(2)^{2x}$. The equation gives the estimated number of online newsletter subscribers at the end of every six-month period, x , starting from the end of January 1997, where $0 \leq x \leq 4$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

(A) The estimated number of online newsletter subscribers at the end of the first six-month period was 500.

(B) The estimated number of online newsletter subscribers at the end of January 1997 was 500.

(C) The estimated number of online newsletter subscribers at the end of January 1998 was 1,000.

(D) The estimated number of online newsletter subscribers at the end of January 1997 was 1,000.

17. The solution to the given system of equations is (x, y) . What is the value of $10x$?

$$2x + 5(y - 14) = 41$$

$$x + 5(y - 14) = 29$$

Student-produced response: enter your answer as a number.

18. When several resistors are connected in series in a circuit, the total resistance is the sum of the resistances. A certain circuit consists of 8 resistors connected in series, where each resistance is positive. The circuit has total resistance 130 ohms. The total resistance of 3 resistors is 90 ohms. Which inequality best represents all possible values of the resistance x , in ohms, of one of the other 5 resistors?

(A) $0 < x < 8$

(B) $0 < x < 40$

(C) $40 < x < 90$

(D) $40 < x < 130$

19. A family has money in an account for renting movies online. Each rental costs the same amount. The function $f(m) = 21 - 3m$ gives the amount of money, in dollars, in the account after the family has rented m movies. Which represents the amount withdrawn each time the family rents a movie?

(A) $3m$

(B) 3

(C) 21

(D) $21 - 3m$

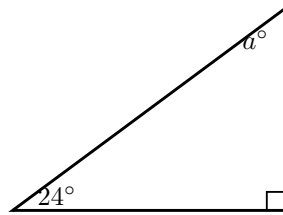
20. If $9 - 4(4 - 9x) = 2 - 5(4 - 9x)$, what is the value of $4 - 9x$?

Student-produced response: enter your answer as a number.

21. On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 21.5% of the farmland and on exactly 14.0% of the pasture. If there are buildings on exactly $p\%$ of the total plot, what is p ?

Student-produced response: enter your answer as a number.

1. In the right triangle shown, what is the value of a ?



- (A) 24 (B) 66
(C) 90 (D) 114
2. The graph of polynomial function f passes through $(5, 3)$. What is $f(5)$?
- (A) 3 (B) 5
(C) 8 (D) 15
3. A circle has a circumference of 42π centimeters. What is its diameter, in centimeters?

Student-produced response: enter your answer as a number.

4. For an object of mass 49 kg, $K = \frac{49}{2}v^2$. Which equation expresses v in terms of K ?

- (A) $\frac{2K}{49}$ (B) $\sqrt{\frac{49}{2K}}$
(C) $\sqrt{\frac{2K}{49}}$ (D) $\frac{2K}{25}$

5. $f(x) = 4x^2$. What is $f(11)$?

- (A) 44 (B) 52
(C) 88 (D) 484

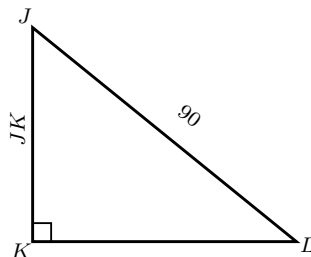
6. The function g is defined by $g(x) = \frac{x+1}{5}$. What is the y -intercept of the graph of $y = g(x)$ in the xy -plane?

- (A) $(0, 0)$ (B) $\left(0, \frac{1}{5}\right)$
(C) $(0, 5)$ (D) $(0, 1)$

7. A city's population doubled every 75 years from 1652 to 1952. It was 160,000 in 1952. What was it in 1652?

Student-produced response: enter your answer as a number.

8. In right triangle JKL , $\tan(L) = \frac{4}{3}$. What is JK ?



- (A) 35 (B) 54
(C) 72 (D) 90

9. $a(7x - 14) + 5a = 7(ax - 2a) - 35$. If the equation has infinitely many solutions, what is a ?

- (A) 0 only
(B) -7 only
(C) 7 only
(D) No real number

10. If $x > 0$ and x is 33% of y , which expression represents y in terms of x ?

- (A) $\frac{100}{33}x$
(B) $\frac{33}{100}x$
(C) $\frac{67}{100}x$
(D) $\frac{133}{100}x$

11. In the equation $(z^m)^{\frac{7}{10}} = \sqrt{z}$, m is a constant and $z > 1$. What is the value of m ?

Student-produced response: enter your answer as a number.

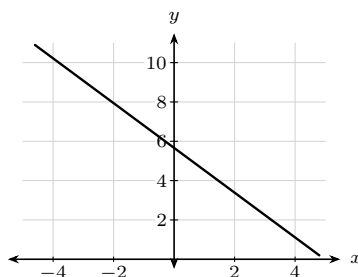
12. A book club charges \$13 for the first book and \$11 for each additional book. Which equation gives total cost y for x books, $x > 0$?

- (A) $y = x + 13$
(B) $y = 11x + 13$
(C) $y = 11(x - 1) + 13$
(D) $y = (x - 1) + 13$

13. $z(w) = (0.829)^{2w}$. $z(w)$ decreases by $p\%$ for each increase of 1 in w . Which is closest to p ?

- (A) 0.171
(B) 0.829
(C) 17.1
(D) 31.3

14. The graph of $y = f(x) + 11$ is a decreasing line. If $c, d > 0$, which equation could define f ?



- (A) $f(x) = -d - cx$
(B) $f(x) = d + cx$
(C) $f(x) = -d + cx$
(D) $f(x) = d - cx$

15. $21x^2 + (21s + r)x + rs = 0$. The product of the solutions is krs . What is k ?

Student-produced response: enter your answer as a number.

16. A is 6,000% of C , and C is 0.008% of B . Which expression equals $a + b$ in terms of c ?

- (A) $12,560c$
(B) $6,125c$
(C) $6,008c$
(D) $1,850c$

17. The quadratic expression $7(x - 5)(x + 3)$ can be rewritten as $ax^2 + bx + c$, where a , b , and c are constants. What is the value of $a + b + c$?

- (A) -112
(B) -56
(C) -28
(D) -7

18. $f(x) = 66b^x$. Each increase of 1 in x increases $f(x)$ by $c\%$, $0 < c < 100$. Which expression gives c ?

(A) $1 + \frac{b}{100}$

(B) $b + 100$

(C) $100(b + 1)$

(D) $100(b - 1)$

19. A tree grows 87 cm every m months. Which expression is its growth every k years?

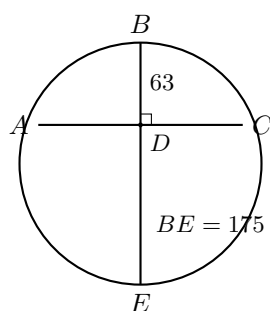
(A) $\frac{29m}{4k}$

(B) $\frac{29k}{4m}$

(C) $\frac{1,044m}{k}$

(D) $\frac{1,044k}{m}$

20. Points A , B , C , and E lie on a circle, and $\overline{BE}$ is a diameter with length 175. Chord $\overline{AC}$ is perpendicular to $\overline{BE}$ at D , and $BD = 63$. What is the length of $\overline{AB}$?

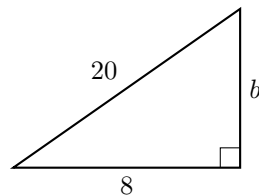


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1. The equation $s = 6t + 18$ gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, 4 seconds after it began to accelerate?

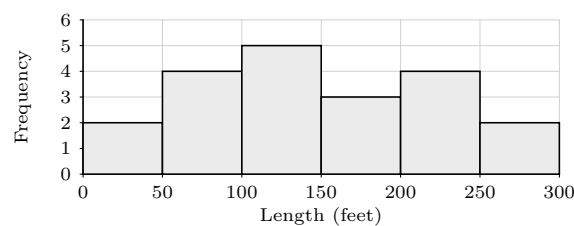
- (A) 40 (B) 42
(C) 48 (D) 54

2. Which equation shows the relationship between the side lengths of the given right triangle?



- (A) $8b = 20$ (B) $8^2 + b^2 = 20^2$
(C) $8^2 + 6^2 = 20^2$ (D) $8^2 - 6^2 = 20^2$

3. The histogram shows 20 lengths, in feet. Which choice could be the median of the data set?



- (A) 69 (B) 119
(C) 169 (D) 219

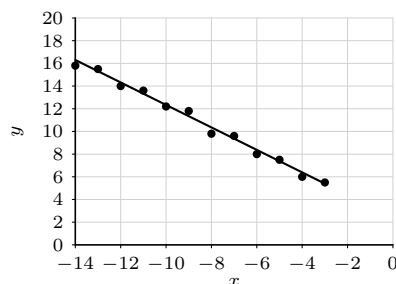
4. A line segment of length 113 cm is divided into three parts. One part is 47 cm. The other two parts have equal lengths. What is the length, in cm, of either equal part?

Student-produced response: enter your answer as a number.

5. $7x + 28 = 7x + k$. The equation has infinitely many solutions. What is k ?

Student-produced response: enter your answer as a number.

6. The scatterplot shows the relationship between x and y and a line of best fit. Which equation best represents the line of best fit?



- (A) $y = -x + 16.3$ (B) $y = x - 2.3$
(C) $y = x + 2.3$ (D) $y = -x + 2.3$

7. The expression $7x^6 + 9x^6 - 8x^6$ is equivalent to bx^6 , where b is a constant. What is b ?

Student-produced response: enter your answer as a number.

8. At what point (x, y) do the equations $y = 6x - 2$ and $2x + y = 14$ intersect in the xy -plane?

- (A) (6, 5) (B) (5, 6)
(C) (2, 10) (D) (10, 2)

9. A rectangle has a length 72 times its width. The function $y = (72w)(w)$ represents its area. What is the best interpretation of $72w$?

- (A) The length, in feet
(B) The area, in square feet
(C) The difference between length and width, in feet
(D) The width, in feet

10. For the linear function $f(x) = 2x + 29$, which table gives three x -values and their corresponding $f(x)$ -values?

(A)

x	0	1	2
$f(x)$	29	31	33

(B)

x	0	1	2
$f(x)$	29	30	31

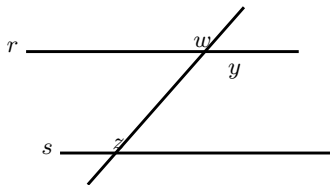
(C)

x	0	1	2
$f(x)$	2	31	60

(D)

x	0	1	2
$f(x)$	29	58	87

11. Line k intersects lines r and s . If $w = 160^\circ$, which additional information is sufficient to prove $r \parallel s$?



- (A) $z = 20^\circ$ (B) $y = 160^\circ$
(C) $w + y = 180^\circ$ (D) $y + z = 180^\circ$

Figure not drawn to scale.

12. In the xy -plane, the circle $(x - 3)^2 + (y + 2)^2 = 100$ is inscribed in a square. What is the perimeter of the square?

- (A) 4 (B) 20
(C) 80 (D) 320

13. A webinar series had 1,250 attendees at the first webinar. 46% attended the second, and 32% of those who attended both first and second attended the third. How many attended all three?

Student-produced response: enter your answer as a number.

14. The exponential function $f(x) = 24(2)^{x/6}$ is given. Which table gives four x -values and their corresponding $f(x)$ -values for the function?

(A)

x	-6	0	6	12
$f(x)$	12	24	48	96

(B)

x	-6	0	6	12
$f(x)$	12	24	36	48

(C)

x	-6	0	6	12
$f(x)$	48	24	12	6

(D)

x	-6	0	6	12
$f(x)$	6	24	96	384

15. Wood samples were cut into cubes. One cube has edge length 3.000 cm and density 0.250 g/cm^3 . What is its mass, in grams?

Student-produced response: enter your answer as a number.

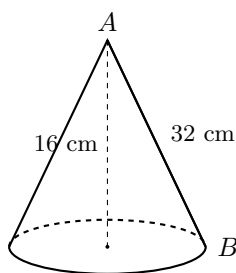
16. $2|5x + 9| + 7 = -11$. What are all solutions?

- (A) 0
(B) 18
(C) 0 and 18
(D) There is no solution

17. What is the sum of the solutions to the equation $\sqrt{2}x^2 - 84\sqrt{2}x + 14 = 0$?

- (A) 0
(B) 7
(C) 42
(D) 84

18. For the right circular cone shown, B lies on the circumference of the base, $AB = 32 \text{ cm}$, height = 16 cm, and volume = $k\pi \text{ cm}^3$. Find k .

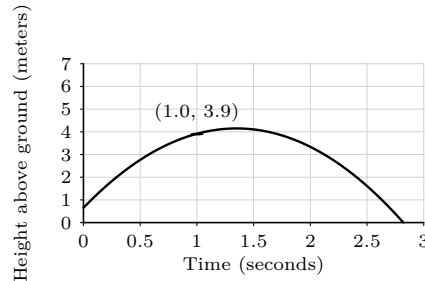


Student-produced response: enter your answer as a number.

19. Lines k and ℓ are perpendicular and intersect at $(2, 8)$. If line k is $y = mx + b$, $m > 1$, which point lies on line ℓ ?

- (A) $\left(3, 8 - \frac{1}{m}\right)$
(B) $\left(3, 8 + \frac{1}{m}\right)$
(C) $(3, 8 - m)$
(D) $(3, 8 + m)$

1. The graph shows the height above ground, in meters, of a ball x seconds after launch. Which statement best interprets the marked point $(1.0, 3.9)$?



- (A) 1.0 second after launch, the height is 3.9 meters.
- (B) 3.9 seconds after launch, the height is 1.0 meter.
- (C) The initial height is 1.0 meter and the initial velocity is 3.9 m/s.
- (D) The ball lands 3.9 meters from where it was launched.

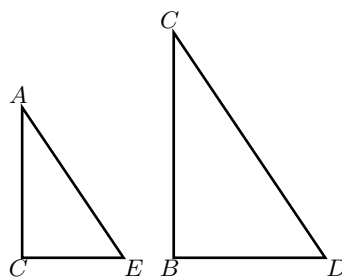
2. In one week, a technician earned \$700. The equation $16h + 13c = 700$ models hours h at the regular job and c at the second job. What does 16 represent?

- (A) Dollars earned per hour at regular job
- (B) Dollars earned per hour at second job
- (C) Hours at regular job
- (D) Hours at second job

3. Which list has the smallest standard deviation?

- (A) 44, 45, 47, 49, 50
- (B) 45, 45, 47, 49, 49
- (C) 45, 46, 47, 48, 49
- (D) 46, 47, 47, 47, 48

4. Triangle CAE is similar to triangle CBD . $m\angle CBD = 56^\circ$, and $AE = 24(BD)$. What is $m\angle CAE$?



- (A) 24°
- (B) 56°
- (C) $(24 + 56)^\circ$
- (D) $(24 - 56)^\circ$

Figure not drawn to scale.

5. A convention center has 375 visitors in rooms A , B , or C . $P(A) = 0.64$ and $P(B) = 0.32$. How many visitors are in room C ?

- (A) 4 (B) 15
(C) 120 (D) 150

6. The solution to the system of equations $x + 2y = 14$ and $3x - y = 7$ is (x, y) . What is the value of $2x + 3y$?

- (A) 26 (B) 20
(C) 14 (D) 23

7. In 5 days, a polar bear ate 22.0 pounds of fat. Which equation describes the amount y , in pounds, the bear ate per day, assuming a constant rate?

- (A) $y = 22 + 5$ (B) $y = 22$
(C) $y = \frac{22}{5}$ (D) $y = 22(5)$

8. A nuclide sample decreases by 22% every 11.11 minutes. Which equation could model mass M after t minutes?

- (A) $M = 100(0.22)^{t/11.11}$ (B) $M = 100(0.22)^{11.11t}$
(C) $M = 100(0.78)^{t/11.11}$ (D) $M = 100(0.78)^{11.11t}$

9. Line ℓ is $y = 8x + 7$. Parallel line j passes through $(0, 15)$. Which equation defines j ?

- (A) $y = 15x + 7$ (B) $y = -15x + 7$
(C) $y = 8x + 15$ (D) $y = -8x + 15$

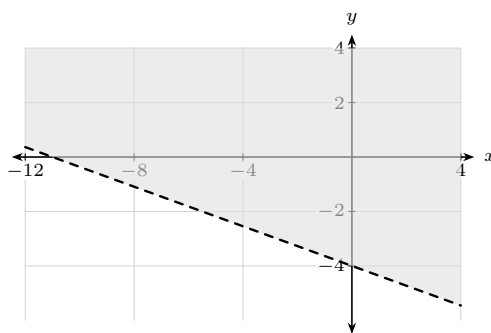
10. For linear p , $p(c) = -6$, $p(5) = 42$, and slope is 8. For linear t , $t(c) = -7$ and $t(6) = 56$. What is the slope of t ?

- (A) -1 (B) 2
(C) 8 (D) 9

11. If $2(4 - 5x) + 7(4 - 5x) + 9 = 8(4 - 5x) + 6$, what is $5x - 4$?

Student-produced response: enter your answer as a number.

12. The shaded region shown represents the solutions to $rx + ty > -44$, where r and t are constants. The boundary line passes through $(-11, 0)$ and $(0, -4)$. What is the value of $r + t$?



- (A) -4 (B) 7
(C) 15 (D) 44

13. The function f is defined by $f(x) = 3x^2$. If $f(a) = 9a$ and $a > 0$, what is the value of a ?

- (A) 1 (B) 3
(C) 9 (D) 27

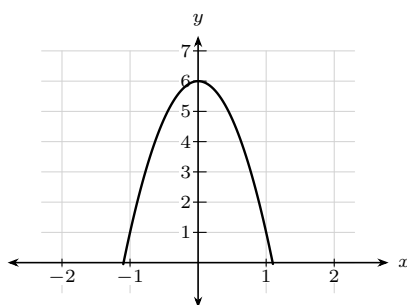
14. In the equation $t^4 = p^{-12}$, $p > 1$ and $0 < t < 1$. If $t = p^{-n}$, where n is a constant, what is the value of n ?

Student-produced response: enter your answer as a number.

15. Quadratic function f has vertex $(1, 6)$ and passes through $(2, 40)$ and $(-1, 142)$. What is the value of $f(-2) - f(0)$?

- (A) 108 (B) 204
(C) 272 (D) 312

16. The graph of $y = f(x) + 2$ is shown. Which equation defines f ?



- (A) $f(x) = -5x^2 + 1$ (B) $f(x) = -25x^2 + 3$
(C) $f(x) = -5x^2 + 4$ (D) $f(x) = -5x^2 + 5$

17. A 45-inch-square map uses 1 inch = 13 miles. A smaller copy has side length 70% shorter. On the smaller map, about how many actual miles does 1 inch represent?

- (A) 3.90 (B) 7.65
(C) 31.50 (D) 43.33

18. In the equation $7x^2(4x - 13)(4x - u) = 0$, u is a positive constant. The sum of the distinct solutions to the equation is 6. What is the value of u ?

Student-produced response: enter your answer as a number.

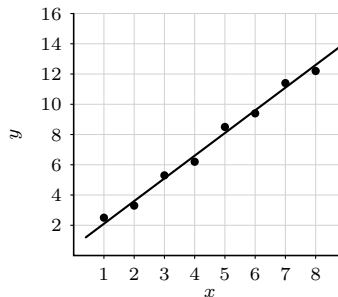
19. Similar right rectangular prisms X and Y have surface areas 59 cm^2 and $1,475 \text{ cm}^2$. Volume of Y is $1,500 \text{ cm}^3$. Find the sum of their volumes.

Student-produced response: enter your answer as a number.

20. $f(x) = ab^{x/n}$, where a, b, n are constants and b, n are integers. If $f(2) = 6$ and $f(5) = 162$, find $f(7)$.

Student-produced response: enter your answer as a number.

1. The scatterplot shows the relationship between x and y . A line of best fit is also shown. Which equation best represents the line of best fit?



- (A) $y = 0.6 - 1.5x$
 (B) $y = 0.6 + 1.5x$
 (C) $y = -0.6 + 1.5x$
 (D) $y = -0.6 - 1.5x$

2. In the system of equations $y = x^2 - 6x + 10$ and $y = 2x - a$, a is a positive integer constant. The system has no real solution. What is the least possible value of a ?

Student-produced response: enter your answer as a number.

3. A rectangle is 8 cm wide. Its length is 50 cm longer than its width. What is its area, in cm^2 ?

- (A) 8
 (B) 16
 (C) 66
 (D) 464

4. The function f is defined by $f(x) = 3(x - 1)$. Which point is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- (A) $(0, -4)$
 (B) $(0, -3)$
 (C) $(0, 0)$
 (D) $(0, 3)$

5. The expression $7x^6 + 9x^6 - 8x^6$ is equivalent to bx^6 . What is b ?

Student-produced response: enter your answer as a number.

6. Which expression is equivalent to $(158y)^{\frac{1}{2}}$, where $y > 1$?

- (A) $158\sqrt{y}$
 (B) $\sqrt{158} - y$
 (C) $\sqrt{158y}$
 (D) $\frac{158}{\sqrt{y}}$

7. A convention center has 275 visitors in rooms A , B , or C . $P(A) = 0.68$ and $P(B) = 0.24$. How many visitors are in room C ?

- (A) 8
 (B) 22
 (C) 121
 (D) 55

8. Line t is defined by $y = 4x + 15$. Line s is perpendicular to line t . What is the slope of line s ?

- (A) 15 (B) 4
(C) -4 (D) $-\frac{1}{4}$

9. The exponential function $f(x) = -10(2)^{x/3}$ is given. Which table gives four x -values and their corresponding $f(x)$ -values for the function?

(A)

x	-3	0	3	6
$f(x)$	-5	-10	-20	-40

(B)

x	-3	0	3	6
$f(x)$	-5	-10	-15	-20

(C)

x	-3	0	3	6
$f(x)$	-20	-10	-5	-2.5

(D)

x	-3	0	3	6
$f(x)$	5	10	20	40

10. $m(x) = 5x + 3$ and $p(x) = 3 - x$. What is $2m(3) - p(3)$?

Student-produced response: enter your answer as a number.

11. Tablets cost \$170 each and 7% tax is added. If a school can spend at most \$53,400 including tax, what is the maximum number of tablets?

Student-produced response: enter your answer as a number.

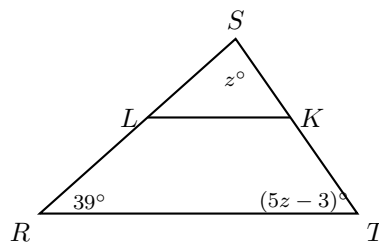
12. The difference between the measures of angle A and angle B is $\frac{3\pi}{5}$ radians. Which expression gives the difference, in degrees?

- (A) $\frac{3\pi}{5} \cdot \frac{180}{\pi}$ (B) $\frac{3\pi}{5} \cdot \frac{\pi}{180}$
(C) $\frac{3\pi}{5} \cdot 180\pi$ (D) $\frac{3\pi}{5} \cdot \frac{360}{\pi}$

13. A nuclide sample decreases by 24% every 22.96 minutes. Which equation could model mass M after t minutes?

- (A) $M = 100(0.76)^{t/22.96}$ (B) $M = 100(0.24)^{t/22.96}$
(C) $M = 100(0.76)^{22.96t}$ (D) $M = 100(0.24)^{22.96t}$

14. In triangle RST , $m\angle R = 39^\circ$, $m\angle S = z^\circ$, and $m\angle T = (5z - 3)^\circ$. L lies on RS , K lies on ST , and $LK \parallel RT$. Find $m\angle SKL$.



Student-produced response: enter your answer as a number.

15. Quadratic function f has vertex $(1, 7)$ and passes through $(2, 53)$ and $(-1, 191)$. What is the value of $f(-2) - f(0)$?

(A) 145

(B) 237

(C) 276

(D) 368

1. In one week, a technician earned \$716. The equation $20h + 14c = 716$ models hours at regular and second jobs. What does 20 represent?

- (A) Dollars per hour at regular job
- (B) Dollars per hour at second job
- (C) Hours at regular job
- (D) Hours at second job

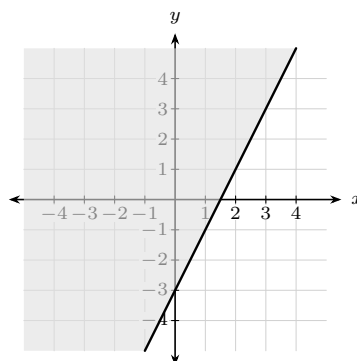
2. A blue whale weighs 5,710 pounds at birth and gains 10.0 pounds per hour. What is its estimated weight 7 hours after birth?

Student-produced response: enter your answer as a number.

3. The equation $34qr = 8s$ relates the positive numbers q , r , and s . Which equation correctly expresses q in terms of r and s ?

- (A) $q = \frac{8s}{34r}$
- (B) $q = \frac{8 - 34r}{s}$
- (C) $q = s + 34r$
- (D) $q = \frac{34rs}{8}$

4. The shaded region shown represents the solutions to $y \geq 2x - 3$. Which ordered pair is a solution to the inequality?



- (A) (0, 5)
- (B) (2, 0)
- (C) (0, -5)
- (D) (3, 2)

5. A rectangle has length 49 times width. $y = (49w)(w)$ gives area. What does $49w$ represent?

- (A) Length in feet
- (B) Area in square feet
- (C) Difference between length and width
- (D) Width in feet

6. If $5(x + 4) = 4(x + 4) + 39$, what is the value of $x + 4$?

- (A) -4
- (B) 35
- (C) 39
- (D) 43

7. The solution to the system of equations $3x + y = 19$ and $x - y = 1$ is (x, y) . What is the value of $2x$?

- (A) 10 (B) 24
(C) 6 (D) 14

8. In 8 days, a polar bear ate 34.4 pounds of fat. Which equation describes the amount y eaten in these 8 days?

- (A) $y = 8$ (B) $y = 4.3$
(C) $y = 34.4$ (D) $y = 34.4 + 8$

9. A wooden cube has edge length 3.000 cm and density 0.220 g/cm^3 . What is its mass, in grams?

Student-produced response: enter your answer as a number.

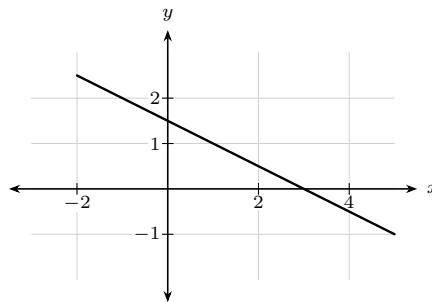
10. What are all solutions to the equation $x^3 + 2x^2 = 0$?

- (A) 0 (B) 0 and -2
(C) 0 and 2 (D) -2 only

11. For linear p , $p(c) = -2$, $p(4) = 38$, slope = 8. For linear t , $t(c) = -3$ and $t(5) = 51$. What is the slope of t ?

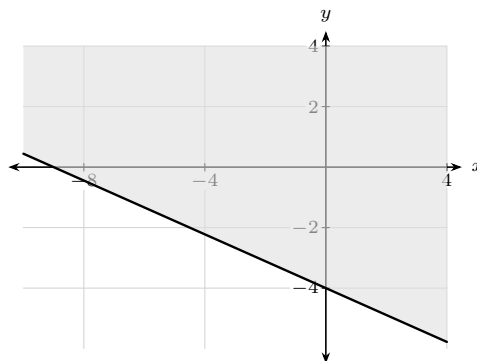
- (A) -1 (B) 6
(C) 8 (D) 9

12. The graph of a linear relationship in the xy -plane is shown. It passes through $(0, 1.5)$ and $(3, 0)$. Which equation represents the graph, where $R > 0$?



- (A) $Rx + 12y = 18$ (B) $Rx - 12y = 18$
(C) $Rx + 12y = -18$ (D) $12x + Ry = 18$

13. The shaded region shown represents the solutions to $rx + ty \geq -36$, where r and t are constants. The boundary line passes through $(-9, 0)$ and $(0, -4)$. What is the value of $r + t$?

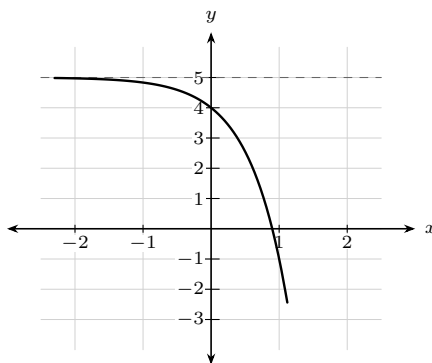


- (A) 13 (B) 5
(C) -13 (D) 36

14. The edge length of cube Y is 3 times the edge length of cube X . The surface area of cube Y is n times the surface area of cube X . What is the value of n ?

- (A) 9 (B) 27
(C) 3 (D) 6

15. The graph of $y = f(x) + 1$ is shown. Which equation defines f ?



- (A) $f(x) = -6^x + 1$ (B) $f(x) = -6^x + 3$
(C) $f(x) = -6^x + 4$ (D) $f(x) = -6^x + 5$

16. Which expression is equivalent to $5(3x + 2)$?

- (A) $15x + 10$ (B) $15x + 2$
(C) $8x + 10$ (D) $15x + 25$

17. A 35-inch-square map uses 1 inch = 13 miles. A smaller copy has side length 70% shorter. On the smaller map, about how many miles does 1 inch represent?

- (A) 3.90 (B) 7.65
(C) 31.50 (D) 43.33

18. In the system of equations $y = -6(x - 12)^2$ and $y = c$, c is a constant and the system has two distinct real solutions. Which value could c be?

- (A) 7 (B) 11
(C) 17 (D) -24

19. In triangle ABC , $\angle A = 58^\circ$ and $AC = 30$. In triangle PQR , $\angle P = 58^\circ$ and $PR = 120$. Which additional information proves similarity?

- (A) $AB = 35$ and $PQ = 35$
(B) $AB = 35$ and $QR = 140$
(C) $\angle B = 34^\circ$ and $\angle R = 88^\circ$
(D) $\angle B = 58^\circ$ and $\angle Q = 34^\circ$

20. An equilateral triangle has perimeter 876 cm and its vertices lie on a circle. The circle radius is $w\sqrt{3}$ cm. Find w .

Student-produced response: enter your answer as a number.

21. The function f is defined by $f(x) = 2^{x/a} - 16$, where $a < 0$. What is the value of $f(4a) \cdot f(9a)$?

Student-produced response: enter your answer as a number.

22. $f(x) = ab^{x/n}$, where a, b, n are constants and b, n are integers. If $f(5) = 7$ and $f(8) = 189$, find $f(10)$.

Student-produced response: enter your answer as a number.

1. The function h is defined by $h(x) = 8|x|$. What is the value of $h(-5)$?

- (A) -40 (B) 3
(C) 13 (D) 40

2. The area of a rectangle is 60 square inches. The length of the longest side of the rectangle is 15 inches. What is the length, in inches, of the shortest side of this rectangle?

- (A) 4 (B) 15
(C) 30 (D) 45

3. For certain air temperatures, the table gives wind chill temperatures for two different wind speeds.

Air temperature	Wind chill temperature at 15 mph	Wind chill temperature at 45 mph
30°F	19°F	12°F
34°F	24°F	18°F
38°F	29°F	23°F

According to the table, what is the wind chill temperature, in degrees Fahrenheit, when the air temperature is 38°F and the wind speed is 15 miles per hour? (Disregard the degree symbol when entering your answer.)

Student-produced response: enter your answer as a number.

4. Charles saves $\frac{2}{5}$ of the \$145 he earns each week from his summer job. If Charles continues to save at this rate, how much money, in dollars, will Charles save in 3 weeks?

Student-produced response: enter your answer as a number.

5. If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

Type	Frequency
Igneous	10
Metamorphic	36
Sedimentary	34

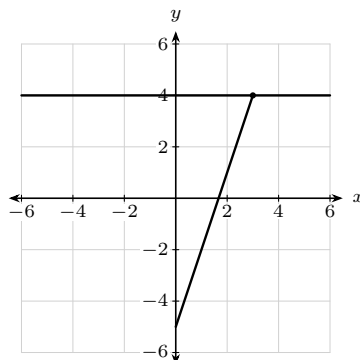
- (A) $\frac{10}{80}$ (B) $\frac{10}{70}$
(C) $\frac{10}{36}$ (D) $\frac{10}{34}$

6. The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

$$\frac{j}{7} = k + 15m$$

- (A) $j = \frac{k + 15m}{7}$ (B) $j = 7k + 15m$
(C) $j = 7(k + 15m)$ (D) $j = k + 7(15m)$

7. The graph of a system of linear equations is shown. What is the solution (x, y) to the system?



- (A) $(4, 4)$
 (B) $(3, 4)$
 (C) $(2, 4)$
 (D) $(0, 4)$

8. A student council group is selling car stickers for a fundraiser. They use the function $p(x) = 5x - 210$ to determine their profit $p(x)$, in dollars, for selling x car stickers. In order to earn a profit of \$800, how many car stickers must they sell?

Student-produced response: enter your answer as a number.

9. If $\frac{x}{y} = 76$ and $\frac{cx}{4y} = 76$, what is the value of c ?

- (A) 4
 (B) 19
 (C) 76
 (D) 304

10. Trapezoid $CDEF$ is similar to trapezoid $JKLM$, such that C, D, E, F correspond to J, K, L, M , respectively. The length of each side of trapezoid $JKLM$ is 3 times the length of its corresponding side in trapezoid $CDEF$. The measure of angle C is 42° . What is the measure of angle J ?

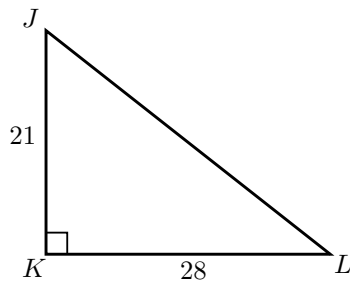
Student-produced response: enter your answer as a number.

11. The table gives the average time t , in minutes, it takes Adriana to travel a certain distance d , in miles. Which equation could represent this linear relationship?

Distance (miles)	Average time (minutes)
0.32	8
0.48	12
0.76	19

- (A) $t = \frac{1}{4}d$
 (B) $t = 25d$
 (C) $t = \frac{1}{25}d$
 (D) $t = 4d$

12. In right triangle JKL , $\tan L = \frac{3}{4}$. What is JK ?



- (A) 16
(B) 35
(C) 15
(D) 21

13. What is the minimum value of the given function?

$$f(x) = x^2 - 12x + 12$$

- (A) -24
(B) -12
(C) 6
(D) 12

14. For the linear function g , $g(9) = 11$ and $g(x+1) - g(x) = 2$. Which equation defines g ?

- (A) $g(x) = 9x + 2$
(B) $g(x) = 9x + 11$
(C) $g(x) = 2x - 18$
(D) $g(x) = 2x - 7$

15. For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Function f is defined by $f(x) = rx + s$, where r and s are constants. What is the value of rs ?

x	0	3	6
$f(x)$	13	52	91

Student-produced response: enter your answer as a number.

16. What are the solutions to the given equation?

$$x(x + 11) - 15 = 0$$

- (A) $x = \frac{11 \pm \sqrt{61}}{2}$
(B) $x = \frac{11 \pm \sqrt{181}}{2}$
(C) $x = \frac{-11 \pm \sqrt{61}}{2}$
(D) $x = \frac{-11 \pm \sqrt{181}}{2}$

17. What is the sum of the solutions to the given equation?

$$2|4x - 2| = 5$$

Student-produced response: enter your answer as a number.

18. The positive number a is 220% of the number b , and a is 70% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

- (A) 154
(B) 171
(C) 214
(D) 314

19. An algae population is 128% greater every 6 hours and has mass 629.60 g after 18 hours. Which model $A(x)$ fits?

- (A) $34.98(1.28)^{x/6}$
(B) $34.98(2.28)^{x/6}$
(C) $53.12(2.28)^{x/6}$
(D) $300.22(1.28)^{x/6}$

20. Which ordered pair (x, y) is a solution to the given system of equations?

$$y = 7x + 3, \quad y = 7x^2 + 3$$

- (A) $(0, 0)$
(B) $(0, 3)$
(C) $(6, 45)$
(D) $(6, 87)$

21. What is an x -intercept of the graph of $y = x^2 - 169$ in the xy -plane?

Student-produced response: enter your answer as a number.

22. A right square pyramid has a height of 15 inches and a volume of 125 cubic inches. What is the perimeter, in inches, of the base of this pyramid?

Student-produced response: enter your answer as a number.

1. For the linear function f , $f(0) = 0$ and $f(40) = 8$. Which equation defines f ?

(A) $f(x) = \frac{x}{8}$

(B) $f(x) = \frac{x}{5}$

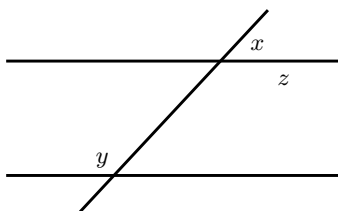
(C) $f(x) = x + 5$

(D) $f(x) = x + 8$

2. The exponential function g is defined by $g(x) = 18a^x$, where a is a positive constant. If $g(3) = 486$, what is the value of $g(4)$?

Student-produced response: enter your answer as a number.

3. Lines n and s are parallel and cut by t . If $x = 6z - 87$ and $y = 3z + 51$, find z .



(A) 14

(B) 24

(C) 46

(D) 57

4. The point $(x, 38)$ satisfies $y > 15$ and $3x + y < 19$. Which could be x ?

(A) 7

(B) 3

(C) -3

(D) -7

5. The function $f(x) = 231(1.026)^x$ models the value, in dollars, of a certain bank account from 1953 through 1973, where x is the number of years after the end of 1953. Which of the following is the best interpretation of $f(5) \approx 263$ in this context?

(A) The value of the bank account is estimated to be approximately 5 dollars greater at the end of 1958 than at the end of 1953.

(B) The value of the bank account is estimated to be approximately 263 dollars at the end of 1958.

(C) The value, in dollars, of the bank account is estimated to be approximately 5 times greater at the end of 1958 than at the end of 1953.

(D) The value of the bank account is estimated to increase by approximately 263 dollars every 5 years between the end of 1953 and the end of 1973.

6. For all values of x , the given expression is equivalent to which of the following?

$$(5x + 10)(2x + 1) - (2x - 18)$$

I. $10x^2 + 23x + 28$ II. $10x^2 + 25x + 10 - 2x - 18$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

7. $y < 4x + 15$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

Table A

(A)

x	y
2	22
4	29
6	35

Table B

(B)

x	y
2	23
4	31
6	39

Table C

(C)

x	y
2	25
4	30
6	36

Table D

(D)

x	y
2	20
4	33
6	38

8. $f(x) = -\frac{1}{4}(x - 9) + 4$. For what x does $f(x) = 2a$?

(A) $-8a + 25$

(B) $-8a - 7$

(C) $-\frac{a}{2} + \frac{25}{4}$

(D) $-\frac{a}{2} + \frac{7}{4}$

9. If $\sqrt{x - 5} = 8$, what is the value of $x - 5$?

(A) 8

(B) 59

(C) 64

(D) 128

10. The solution to the given system of equations is (x, y) . What is the value of $8x + 6y$?

$$2(8x) + 7(6y) = 21, \quad -2(8x) + 7(6y) = 21$$

(A) 0.5

(B) 1.5

(C) 3

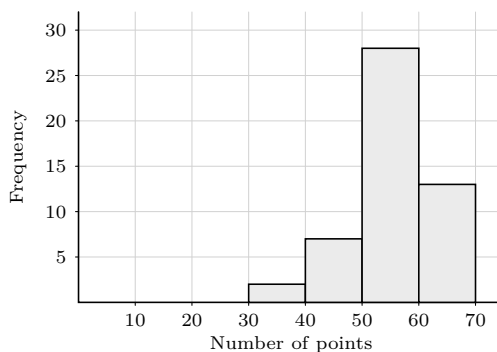
(D) 21

11. The given expression, where b is a constant, is equivalent to $(3x^9 + q)(rx^9 + 2)$, where q and r are constants. What is the value of b ?

$$57x^{18} + bx^9 + 34$$

Student-produced response: enter your answer as a number.

12. The histogram summarizes the distribution of an original data set that represents the number of points per game a basketball team has scored in the last 50 games played. If the team scores 17 points in the next game and this game is added to the original data set to create a new data set of 51 values, which of the following must be true?



I. The median number of points per game for the new data set is less than the median number of points per game for the original data set.

II. The mean number of points per game for the new data set is less than the mean number of points per game for the original data set.

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

13. $g(x) = x^3 + ax^2 + bx + c$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 3 . What is the value of a ?

(A) -42

(B) -6

(C) 6

(D) 42

14. $f(x) = 18(2.10)^{x/2}$

The function f is defined by the given equation. The value of $f(x)$ increases by $p\%$ for every increase of x by 4 . For which of the following functions, where n is a positive constant, does the value of $g(x)$ increase by $p\%$ for every increase of x by 1 ?

(A) $g(x) = n(1.05)^x$

(B) $g(x) = n(2.10)^x$

(C) $g(x) = n(3.41)^x$

(D) $g(x) = n(4.41)^x$

15. Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 3.1 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

(A) $0.83x + 0.33y = 3.1$

(B) $3.0x + 1.2y = 3.1$

(C) $1.2x + 3.0y = 3.1$

(D) $0.33x + 0.83y = 3.1$

16. A circle in the xy -plane has its center at $(-4, 4)$ and has a radius of 6 . An equation of this circle is

$$x^2 + y^2 + ax + by + c = 0,$$

where a , b , and c are constants. What is the value of c ?

Student-produced response: enter your answer as a number.

17. The right rectangular prism X is similar to the right rectangular prism Y . The surface area of right rectangular prism X is 62 square centimeters, and the surface area of right rectangular prism Y is 1550 cm^2 . The volume of the right rectangular prism X is 16.8 cubic centimeters. What is the volume, in cm^3 , of the right rectangular prism Y ?

Student-produced response: enter your answer as a number.

18. Predator and prey populations start equal. Prey increases by 2100% and predators by 450% . Predators end $p\%$ less than prey. Find p .

Student-produced response: enter your answer as a number.

19. The area of a rectangular region is increasing at a rate of 220 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter $= 3.28$ feet.)

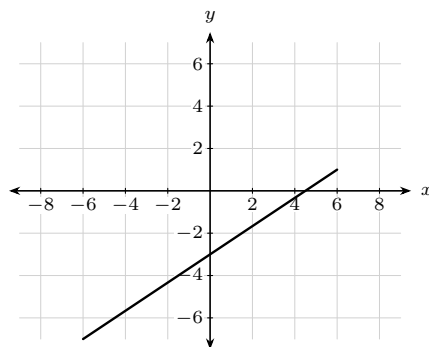
(A) 0.34

(B) 1.12

(C) 12.03

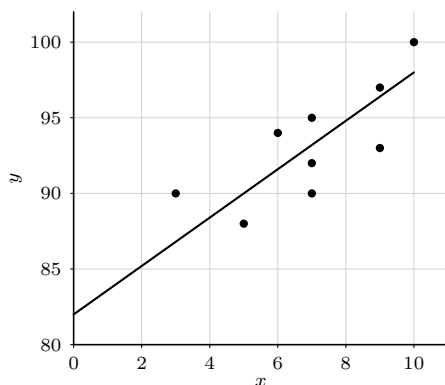
(D) 20.45

- 20.** Line k is shown in the xy -plane. Line n (not shown) is perpendicular to line k and passes through the point $(2, 7)$. Which equation defines line n ?



- (A) $y = -\frac{3}{2}x + 4$ (B) $y = -\frac{3}{2}x + 10$
 (C) $y = \frac{2}{3}x + \frac{17}{3}$ (D) $y = \frac{3}{2}x + 4$

- 21.** The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown. For how many of the 9 data points is the actual y -value greater than the y -value predicted by the line of best fit?



- (A) 2 (B) 5
 (C) 7 (D) 8

- 22.** If $9(8 - 3x) + 2 = 8(8 - 3x) + 19$, what is $8 - 3x$?

- (A) -17 (B) -3
 (C) 17 (D) 3

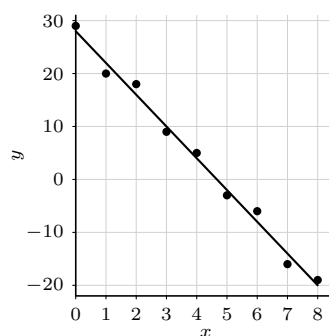
1. A table summarizes the UV index recorded on each of 40 days. On how many days was UV index 1 recorded?

UV index	Number of days
1	9
2	15
3	12
4	13

- (A) 4 (B) 9
(C) 15 (D) 40
2. A woodworker charged \$17 per hour plus a one-time \$45 fee. The equation $17h + 45 = 164$ models the charge. What does 164 represent?
- (A) One-time fee (B) Hours worked
(C) Hourly charge (D) Total charge
3. Recorded temperatures ranged from 40°F to 90°F . Which inequality is true for every recorded t ?
- (A) $40 \leq t \leq 90$ (B) $t \leq 40$
(C) $t < 50$ (D) $t \geq 90$
4. $f(x) = 6(2x + 4)$. For what x is $f(x) = 48$?
- (A) 2 (B) 4
(C) 6 (D) 8
5. The ratio of green to blue tiles is 5 : 2. If there are 16 blue tiles, how many green tiles?

Student-produced response: enter your answer as a number.

6. Which equation is the most appropriate linear model for the scatterplot?

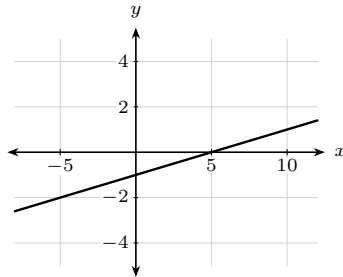


- (A) $y = -6x + 28$ (B) $y = 6 - 28x$
(C) $y = 28 + 6x$ (D) $y = 28 - 6x$

7. A 900 g soil sample contains 189 g of water. What percent of the sample is water?

- (A) 9% (B) 9.9%
(C) 21% (D) 4.76%

8. Line k is perpendicular to line n shown. What is the slope of line k ?



- (A) -5 (B) $-\frac{1}{5}$
(C) $\frac{1}{5}$ (D) 5

9. The graphs of $mx + ky = -83$ and $2x + ky = -22$ intersect at $(5, y)$ in the xy -plane. What is the value of m ?

Student-produced response: enter your answer as a number.

10. A line passes through $(0, 2)$ and $(3, 3)$. What is its slope?

Student-produced response: enter your answer as a number.

11. A 250-page book has 21 illustrated pages. What is the probability a randomly selected page has an illustration?

Student-produced response: enter your answer as a number.

12. $b^2 + 5c = 9d$, with $d > c$. Which equation expresses b ?

- (A) $b = \sqrt{9d + 5c}$ (B) $b = 9d - 5c$
(C) $b = \pm\sqrt{9d + 5c}$ (D) $b = \pm\sqrt{9d - 5c}$

13. A cylinder has a height of 4 m and a base radius of 18 m. What is the volume of the cylinder?

- (A) 72π (B) 288π
(C) $1,296\pi$ (D) $5,184\pi$

14. What is the y -intercept of $3x + 2y = 96$?

- (A) $(0, 5)$ (B) $(0, 6)$
(C) $(0, 32)$ (D) $(0, 48)$

15. If $p + 12 = 54$, find $7p$.

Student-produced response: enter your answer as a number.

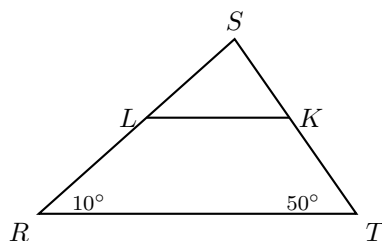
16. A neighborhood population was 1,340 in 2009 and increased approximately 3% each year. Which equation models N after t years?

- (A) $N = 1,340(0.03)^t$ (B) $N = 1.03(1,340)^t$
 (C) $N = 0.03(1,340)^t$ (D) $N = 1,340(1.03)^t$

17. Which of the following points is NOT a solution to the inequality $y > 2x - 3$?

- (A) (0, 0) (B) (1, 0)
 (C) (2, 2) (D) (3, 3)

18. In triangle RST , $\angle R = 10^\circ$ and $\angle T = 50^\circ$. L lies on RS , K on ST , and $LK \parallel RT$. Find $\angle SKL$.



Student-produced response: enter your answer as a number.

19. An auditorium's revenue, in dollars, is modeled by $R(x) = -5x^2 + 240x + 1000$, where x is the increase, in dollars, in the ticket price. At what ticket-price increase is the revenue maximized?

- (A) 8 (B) 24
 (C) 32 (D) 40

20. Which point lies on $(x + 3)^2 + (y - 4)^2 = 25$?

- (A) $(-3, 4)$ (B) $(3, -4)$
 (C) $(\sqrt{10}, \sqrt{14} - 4)$ (D) $(\sqrt{11} - 3, \sqrt{14} + 4)$

21. The expression $\sqrt[3]{x^2} \cdot x^{\frac{10}{3}}$ is equivalent to x^c , where $x > 0$. What is the value of c ?

- (A) 4 (B) 12
 (C) $\frac{2}{3}$ (D) 7

22. Triangles ABC and DEF are similar right triangles, where A corresponds to D and $\angle B$ and $\angle E$ are right angles. $AB = 2.4(DE)$ and $\tan A = \frac{3}{4}$. What is the value of $\sin D$?

Student-produced response: enter your answer as a number.

1. A bird's speed was measured as 49 kilometers per hour. What is the bird's speed, in miles per hour, to the nearest tenth? (Use 1 mile = 1.6 km.)

- (A) 30.6 (B) 47.4
(C) 50.6 (D) 78.4

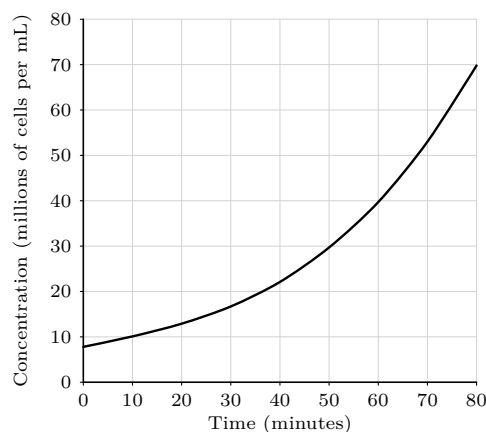
2. Which expression is equivalent to $(3x^3 - x^2 + 4)(5x^2 + 8x)$?

- (A) $15x^5 + 19x^4 - 8x^3 + 20x^2 + 32x$ (B) $15x^5 + 19x^4 - 8x^3 - 20x^2 + 32x$
(C) $15x^5 + 29x^4 - 8x^3 + 20x^2 + 32x$ (D) $15x^5 + 19x^4 + 8x^3 + 20x^2 + 32x$

3. If $5(x + 1) = 50$, what is $x + 1$?

- (A) 5 (B) 10
(C) 15 (D) 20

4. The graph shows the concentration of bacteria in a sample over time. At what time, in minutes, is the concentration 35 million cells/mL?



- (A) 25 (B) 40
(C) 55 (D) 70

5. If $(x + 3)^2 = 48$, what is the value of $x^2 + 6x$?

- (A) 30 (B) 39
(C) 48 (D) 57

6. How many solutions does $y = 5x$ and $y = 2x + 2$ have?

- (A) Exactly one (B) Exactly two
(C) Infinitely many (D) Zero

7. The function f is defined by $f(x) = 171 - 16x$. For which value of k does $f(k) = 3k$?

- (A) 3 (B) 6
(C) 9 (D) 12

8. $g(x) = 2(16x - 17)$. What is the y -coordinate of the y -intercept of $y = g(x) - 3$?

- (A) -37 (B) -34
(C) -20 (D) -17

9. A table gives a linear relationship between x and y in terms of n . What is the slope?

x	y
0	n
4	$n + 19$
8	$n + 38$

Student-produced response: enter your answer as a number.

10. $18x^2 + 24x + c = 0$ has exactly one solution. Find c .

Student-produced response: enter your answer as a number.

11. $(x - 3) - 8(y + 9) = 129$ and $(x - 3) + 8(y + 9) = 432$. Find $8(x - 3)$.

Student-produced response: enter your answer as a number.

12. A square-based rectangular prism has a volume of $2,448 \text{ cm}^3$ and a base area of 144 cm^2 . What is the height, in cm, of the prism?

- (A) 17 (B) 20
(C) 24 (D) 34

13. The system of equations $y = 5(x - 2)^2$ and $y = 10(x - 2) + c$, where c is a constant, has the solution $(3, 5)$. What is the value of c ?

- (A) -5 (B) 5
(C) 10 (D) 15

14. A quadratic has x -intercepts -39 and p , and maximum at $x = 14$. Find p .

Student-produced response: enter your answer as a number.

15. The solution to the equation $3x - k = 87$ is $x = 174$, where k is a constant. What is the value of k ?

Student-produced response: enter your answer as a number.

16. A triangle has area x^2 , base $2x + 6$, and height $x - 2$. Find x .

Student-produced response: enter your answer as a number.

17. Quadratic g has vertex $(-1, -4)$ and passes through $(-2, -43)$ and $(1, -160)$. Find $g(0) - g(2)$.

- (A) -121 (B) 0
(C) 117 (D) 312

18. Congruent triangles ABC and DEF have $A \leftrightarrow D$ and B, E right angles. If $\angle A = 62^\circ$, find $\angle F$.

- (A) 28
(B) 62
(C) 90
(D) 1

19. A consultant charges \$408 for the first hour and \$204 each additional hour. Which $C(h)$ models $h \geq 1$?

- (A) $204h + 204$
(B) $204h + 408$
(C) $408h + 204$
(D) $408h + 612$

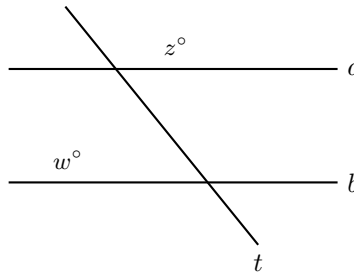
20. A model estimates the noise level, in decibels, at a distance of d meters from a speaker by $N(d) = 47 - 0.4d$. Which is the best interpretation of 47 in this context?

- (A) The estimated noise level at 0 meters
(B) The estimated decrease in noise level for each meter
(C) The estimated noise level at 47 meters
(D) The distance at which the noise level reaches 0

21. For each real number r , which of the following points lies on the graphs of both $y = rx + 5$ and $y = 2rx + 5$ in the xy -plane?

- (A) $(0, 5)$
(B) $(5, 0)$
(C) $(r, 5)$
(D) $(1, 5 + r)$

22. Lines a and b are parallel and are intersected by transversal t . If $z = 49$, which of the following must be true?



- (A) $z + w = 90$
(B) $z + w = 180$
(C) $w = z$
(D) $w = 2z$

SET 6 · QUESTION BANK

A1. $f(x) = 5x^2 - 6$. Find $f(10)$.

Student-produced response: enter your answer as a number.

A2. Wind speeds ranged from 5 to 14 mph. Which inequality holds for every w ?

- (A) $w < 5$ (B) $w \leq 9$
(C) $w > 14$ (D) $5 \leq w \leq 14$

A3. Solve $4p - 5p = 14$.

- (A) -15 (B) -14
(C) 14 (D) 70

A4. A 22-sided polygon has perimeter 131 in. One side is 5 in and the other 21 sides are equal. Find an equal side.

Student-produced response: enter your answer as a number.

A5. $13,000 = 2,600 + 260t$ models a car purchase. What does 260 represent?

- (A) Down payment
(B) Each monthly payment
(C) Number of payments
(D) Total paid after t payments

A6. If $3 - 2x = 42$, what is the value of $6(3 - 2x)$?

- (A) 6 (B) 36
(C) 42 (D) 252

A7. The ratio $72:m$ is equal to the ratio $4:24$. What is the value of m ?

- (A) 12 (B) 24
(C) 144 (D) 432

A8. Solve $(x + 4)(x - 6) = 0$.

- (A) -4 and -6 (B) -4 and 6
(C) 4 and -6 (D) 4 and 6

A9. A solution to $7x - 8y = 12$ and $7x - 8y = 36$ is (x, y) , $x > 0$. Find x .

Student-produced response: enter your answer as a number.

A10. A table shows four values of x and their corresponding values of y .

x	-6	-3	3	6
y	48	39	21	12

Which equation represents the relationship?

- (A) $9x + 3y = 90$ (B) $9x + 3y = 3$
(C) $3x + 9y = 90$ (D) $3x + 9y = 3$

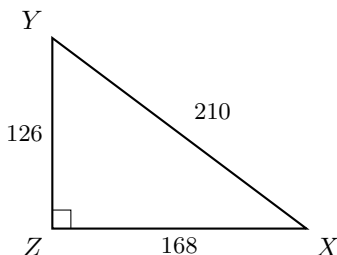
A12. 3,125 attended webinar 1; 56% attended webinar 2; 26% of those attended webinar 3. How many attended all three?

Student-produced response: enter your answer as a number.

A13. A scatterplot has 9 points; 5 are above the line of best fit. How many actual y -values exceed predicted y -values?

- (A) 8 (B) 7
(C) 5 (D) 4

A14. Right triangle XYZ has $XY = 210$, $YZ = 126$, $ZX = 168$, right angle Z . Find $\tan X$.



Student-produced response: enter your answer as a number.

A16. Circle R : $(x + 8)^2 + (y + 19)^2 = 100$. Circle S is R shifted right 3. If S is $(x + h)^2 + (y + k)^2 = 100$, find h .

Student-produced response: enter your answer as a number.

A17. A video has 237 views initially and grows 70% every 2 days. Which model gives views after x days?

- (A) $237(0.70)^{x/2}$ (B) $237(0.70)^x$
(C) $237(1.70)^{x/2}$ (D) $237(1.70)^x$

A18. Triangle TUV is dilated by a scale factor of 2 to form triangle $T'U'V'$. If $\angle V = 26^\circ$, what is the measure of $\angle V'$?

- (A) 26° (B) 52°
(C) 13° (D) 104°

A19. $f(x) = 4x^2 + 16x + 22$ and $h(x) = f(x) + 9$. What is the minimum value of h ?

- (A) 3 (B) 15
(C) 24 (D) 6

A20. The system $7(5x) + 9(8y) = 189$ and $7(5x) - 9(8y) = -819$ has solution (x, y) . What is the value of $5x + 8y$?

- (A) 101 (B) 26
(C) 11 (D) 56

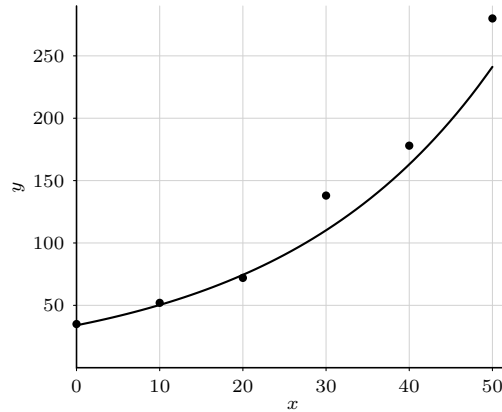
A21. When $f(x)$ is divided by $x - 8$, the remainder is 9. What point must be on $y = f(x)$?

- (A) $(9, -8)$ (B) $(-8, 9)$
(C) $(9, 8)$ (D) $(8, 9)$

B3. A triangle has base 80 cm and height 40 cm. Find its area.

Student-produced response: enter your answer as a number.

B5. Which exponential model best fits the scatterplot shown?



(A) $y = 34(1.04)^{-x}$

(B) $y = 34(1.04)^x$

(C) $y = 34(0.04)^x$

(D) $y = 34 + 1.04x$

B6. A \$70 gift card buys two \$6.50 movies; the rest rents movies at \$1.50 each. How many rentals?

Student-produced response: enter your answer as a number.

B7. An isosceles triangle has vertex angle 138° . Find either base angle.

(A) 89

(B) 69

(C) 42

(D) 21

B8. For $50x + 49y = c$ and $49x - 50y = c$, with $c > 0$, which point could be the intersection?

(A) $(10, -19)$

(B) $(10, 990)$

(C) $(c, 0.5)$

(D) $(c, 0)$

B9. A line has slope 25 and passes through $(0, 0)$. Which function defines it?

(A) $-\frac{x}{25}$

(B) $-25x$

(C) x

(D) $25x$

B10. 375 visitors: $P(A) = 0.48$, $P(B) = 0.16$. How many are in C ?

(A) 29

(B) 36

(C) 120

(D) 135

B11. What is the range of the data set 4, 4, 12, 25, 25?

(A) 12

(B) 14

(C) 21

(D) 29

B13. If $7(x - 3)^2 = 252$, what is the value of $x^2 - 6x$?

- (A) 18 (B) 27
(C) 36 (D) 45

B14. Give one solution to $x^2 - 5x + 2 = 0$.

Student-produced response: enter your answer as a number.

B15. A circle has center $(-7, 2)$ and contains $(1, -2)$. Which equation represents it?

- (A) $(x + 7)^2 + (y - 2)^2 = 80$ (B) $(x + 7)^2 + (y - 2)^2 = 36$
(C) $(x - 7)^2 + (y + 2)^2 = 80$ (D) $(x - 7)^2 + (y + 2)^2 = 36$

B16. A dance studio charges an introductory fee for 3 lessons plus a fixed amount per additional lesson. Four lessons cost \$60 and sixteen cost \$300. Which function gives total charge for $x > 2$?

- (A) $15x$ (B) $15x + 60$
(C) $20x$ (D) $20x - 20$

B17. A dealership has 30% sedans and 50% SUVs. If 57 sedans are for sale, how many minivans?

Student-produced response: enter your answer as a number.

B18. A video has 236 views initially and grows 60% every 3 days. Which model gives views after x days?

- (A) $236(0.60)^{x/3}$ (B) $236(0.60)^x$
(C) $236(1.60)^{x/3}$ (D) $236(1.60)^x$

B19. $0.10x + 0.20y = 0.18(x + y)$. If $x = 45.0$ gallons, find y .

Student-produced response: enter your answer as a number.

B20. $f(x) = x + 3$ and $g(x) = 7x^2 - rx + 63$. If $f(x)g(x) = 7x^3 + 189$, find r .

Student-produced response: enter your answer as a number.

B21. For which of the following tables are all the pairs (x, y) solutions to $y < 14x + 12$?

Table A

(A)

x	y
1	25
2	39
3	53

Table C

(C)

x	y
1	20
2	41
3	50

Table B

(B)

x	y
1	26
2	40
3	54

Table D

(D)

x	y
1	30
2	35
3	60

C1. $h(x) = 100x - 80$. Find $h(4)$.

- (A) 480 (B) 320
(C) 184 (D) 24

C2. A coral is 210 cm initially and grows 8 cm/year. Find its length after 10 years.

- (A) 80 (B) 218
(C) 290 (D) 2,100

C3. 65 hamburger buns are needed; packages contain 12. Find the minimum packages.

Student-produced response: enter your answer as a number.

C4. Solve $8p - 9p = 8$.

Student-produced response: enter your answer as a number.

C5. A soccer team earns 1 point per draw and 3 per win, total 46. Which equation?

- (A) $d + w = 46$ (B) $3d + 4 = 46$
(C) $3d + w = 46$ (D) $d + 3w = 46$

C6. The table gives values of y for $x = 1, 2, 3, 4$. It describes what kind of function?

x	1	2	3	4
y	4000	2400	1440	864

- (A) Decreasing exponential (B) Increasing exponential
(C) Increasing linear (D) Decreasing linear

C8. $f(x) = 47,000(1.04)^x$ models population x years after 1982. Interpret $f(6) \approx 59,470$.

- (A) Increase by 59,470 every 6 years
(B) Population is about 59,470 six years after 1982
(C) Decrease by 59,470 at year 6
(D) Decrease by 59,470 every 6 years

C10. Similar triangles have $AB = 130$, $BC = 650$, and corresponding $YZ = 70$. Find XY .

- (A) 14 (B) 70
(C) 144 (D) 182

C11. $f(x) = 2x + b$ and $f(5) = 4$. Find b .

Student-produced response: enter your answer as a number.

C12. Which expression equals $2(x - 2)(x + 8)$?

- (A) $2x^2 - 72$ (B) $2x^2 + 12x - 32$
(C) $2x^2 - x - 36$ (D) $x^2 + 7x - 36$

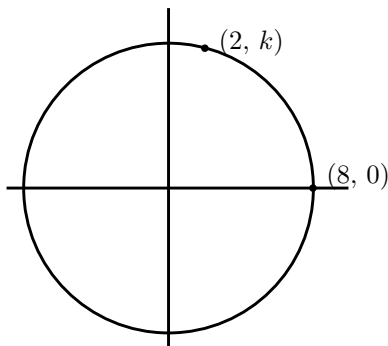
C15. Using a random sample of 230 instead of 115 students most likely has what effect?

- (A) Lower margin of error (B) Higher margin of error
(C) Lower reported mean (D) Higher reported mean

C16. Find the greatest solution of $x^2 - 5x - 5 = 0$.

Student-produced response: enter your answer as a number.

C18. A circle centered at $(0, 0)$ contains $(8, 0)$ and $(2, k)$, $k > 0$. Find k .



(A) 2

(B) 7

(C) 8

(D) $\sqrt{60}$

C19. For acute X, Y , $Y = 24^\circ$ and $\sin X = \cos Y$. Find X .

(A) 21°

(B) 45°

(C) 66°

(D) 69°

C20. 0.1197 is 0.09% of b , and b is 25% of c . Find c .

Student-produced response: enter your answer as a number.

C21. A prism has height 24, base length x , and width $x - 10$. Which volume function?

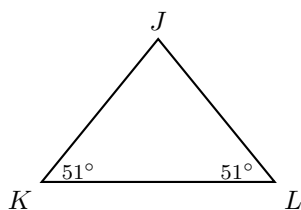
(A) $x(x + 24)(x + 10)$

(B) $x(x + 24)(x - 10)$

(C) $24x(x + 10)$

(D) $24x(x - 10)$

H1. In triangle JKL , $\angle K = \angle L = 51^\circ$. Find $\angle J$.



Student-produced response: enter your answer as a number.

H2. For $y = 7(2)^x + 8$, a table contains the values shown. Find a .

x	1	2	3
y	22	36	a

(A) 120

(B) 64

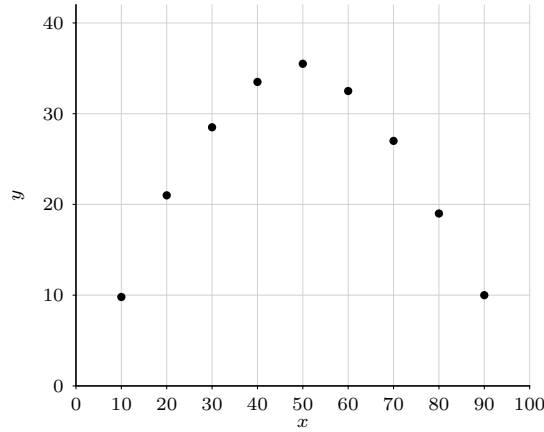
(C) 56

(D) 5

H3. A linear graph lies entirely in quadrants I and II. Which function could define the graph?

- (A) $h(x) = 18x - 36$ (B) $h(x) = 16x$
(C) $h(x) = 16$ (D) $h(x) = -x - 16$

H4. Which quadratic function best models the downward-opening scatterplot shown, whose vertex is near $(50, 35)$?



- (A) $y = -0.007(x - 50)^2 + 35$ (B) $y = -0.007(x - 35)^2 + 50$
(C) $y = 0.007(x - 50)^2 + 35$ (D) $y = -0.007(x + 50)^2 + 35$

H5. Using sample size 180 instead of 90 most likely does what?

- (A) Lowers margin of error (B) Raises margin of error
(C) Lowers mean (D) Raises mean

H7. If $16 + (7x - 6) = 3(7x - 6)$, find $7x - 6$.

Student-produced response: enter your answer as a number.

H8. Solve $8x - 7y = 5$ and $4x + y = 8$; find $x + y$.

Student-produced response: enter your answer as a number.

H9. $H = 2.419L + 22.83$ relates femur length L to height H . Interpret 2.419.

- (A) Height increase per 1-inch femur increase
(B) Femur increase per 1-inch height increase
(C) Femur increase per 22.83-inch height increase
(D) Estimated height for femur L

H10. $0.10x + 0.20y = 0.17(x + y)$. If $x = 75.0$ gallons, find y .

Student-produced response: enter your answer as a number.

H12. Line h passes through $(0, 0)$ and is perpendicular to $-3x + y = 20$. If h contains $(18, t)$, find t .

- (A) -54 (B) -6
(C) 6 (D) 54

H13. Which system has no solution?

- (A) $-3x + 9y = -2$; $3x - 9y = 2$ (B) $3x - 9y = 2$; $x + 10y = 3$
(C) $3x - 9y = 2$; $-9x + 27y = -6$ (D) $3x + 9y = 2$; $6x - 18y = 4$

H14. The function $f(x) = 6000(1.003)^{2x}$ models the balance, in dollars, of a bank account x years after it was opened. Which is the best interpretation of 1.003 in this context?

- (A) Every 2 years the balance rises by \$3
(B) Every 6 months the balance rises by 0.3%
(C) Every 2 years the balance rises by 0.3%
(D) Every year the balance rises by 0.3%

H16. $5|x - 8| = k$ has exactly one solution. Which k could work?

- (A) 11 (B) -8 only
(C) 0 only (D) 8 only

H18. A seal depth is quadratic, with maximum 382.2 m at $t = 7$ and surface at $t = 14$. Estimate depth at $t = 10$.

Student-produced response: enter your answer as a number.

H19. $f(x) = 11x^{10} + 3x^8$ and $g(x) = -17x^7 + 9x^5$. In $p = fg$, find coefficient of x^{15} .

- (A) 15 (B) 18
(C) 99 (D) 150

H20. Chemical masses start equal. A rises 2600%; B rises 380%. About how many times greater is A than B ?

- (A) 29.80 (B) 22.20
(C) 6.84 (D) 5.63

H22. An isosceles right triangle has perimeter $34 + 34\sqrt{2}$. Find one leg.

- (A) 4 (B) $17\sqrt{2}$
(C) 34 (D) $34\sqrt{2}$